Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)
MAKE UP EXAMINATIONS, DECEMBER 2024

SOLID STATE DRIVES [ELE 4088]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: DECEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. A DC-DC converter suitable for 4-quadrant operation is required in an industry. Suggest a suitable circuit configuration and explain its operation. **(02)**

1B. Consider a single-phase controlled rectifier fed 200V, 12A, 900 rpm seperately excited DC motor. The motor has an armature resistance of 1.2 Ω. The converter is supplied with a 230V, 50Hz single-phase AC supply. Calculate the speed at which it runs at a firing angle of 60 degrees.

(Enough filter inductance is added to ensure continuous conduction). Also, draw the waveforms of armature voltage and current in this case. **(04)**

1C Differentiate the v/f and flux weakening control strategies of AC drives with suitable schematic diagrams and suitable speed ranges of operation. **(04)**

2A Is a 10kW motor suitable for the application given below? Justify your answer

A motor needs to be used on a construction site, to drive two loads. One load has rotational motion. It is coupled to the motor through a reduction gear with a gear ratio of 0.1 and an efficiency of 80%. The load has a moment of inertia of 8 kgm^2 and a torque of 8 Nm. Another load has translational motion and consists of 1000 kg to be lifted up at a uniform speed of 1.4 m/s. The coupling between the motor and load has an efficiency of 86%. Assume that, the motor selected has a moment of inertia of 0.22 kgm^2 and has to run at a constant speed of 1410 rpm.

Also, determine the equivalent inertia referred to the motor shaft. **(05)**

2B. Explain the procedure to check whether the armature current is continuous or not, in a single-phase controlled rectifier fed separately excited DC motor using the expression for armature current given below.

$$i_{(\omega t)} = \frac{V_m}{Z}\left[\sin(\omega t - \varphi) - \sin(\alpha - \varphi)e^{((\alpha-\omega t)/\omega\tau)}\right] - \frac{k\omega_m}{R_a}\left[1 - e^{((\alpha-\omega t)/\omega\tau)}\right])$$

(03)

2C. With the help of a suitable block schematic explain the need for DC-DC converters in an Electric Vehicle **(02)**

3A. Draw the waveforms of armature voltage and current of a three-phase controlled rectifier fed separately excited DC motor at a firing angle of 30 degrees operating **(04)**

in discontinuous conduction. Clearly show the triggering sequence. The given template in Fig. Q 3A may be used.

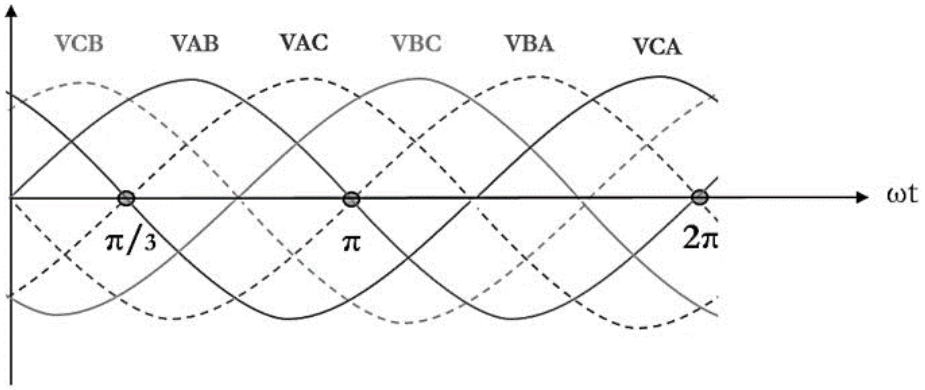


Fig. Q 3A

3B. With the help of speed-torque characteristics and suitable block schematic, differentiate the concept, advantages and disadvantages of rotor resistance control and slip power recovery scheme of a three-phase induction motor. **(04)**

3C. Write a note on the government support schemes in India for Electric vehicles. **(02)**

4A. Demonstrate the switching scheme of a class C-chopper fed separately excited DC motor with the help of the waveforms of armature voltage and current. **(04)**

4B. Write a technical note on EV batteries mentioning the types and emerging technologies for charging the batteries. **(02)**

4C. Discuss the concept of field-oriented control strategy in AC drives with the help of a suitable block schematic. Mention the challenges. **(04)**

5A. Is the operating point 'A' given in the following figure (Fig.5A) steady state stable? Justify your answer. ('T_L' represents load torque and 'Tm' motor torque)

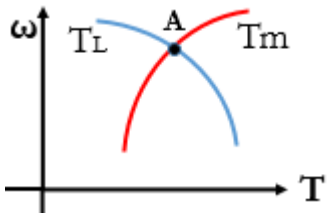


Fig.5A

(03)

5B. Differentiate switched reluctance motor and synchronous reluctance motor based on construction, working principle and distinct features. **(04)**

5C. What are the different components of load torque in a drive system? Discuss the nature of each component with the necessary characteristics. **(03)**